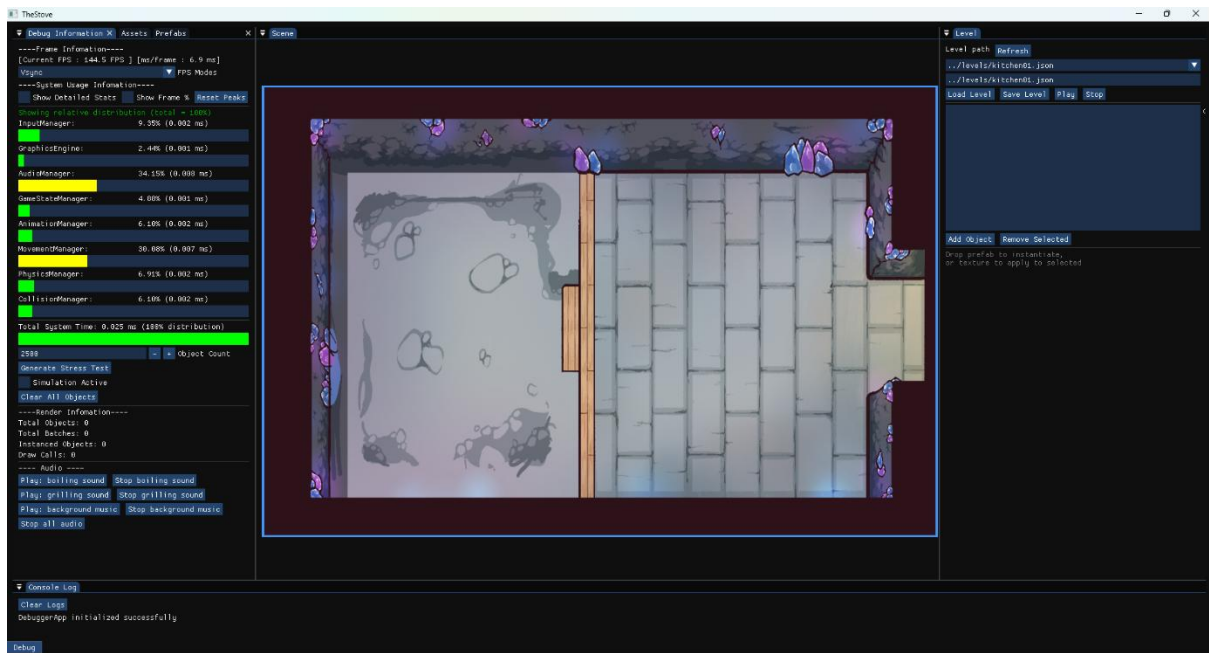
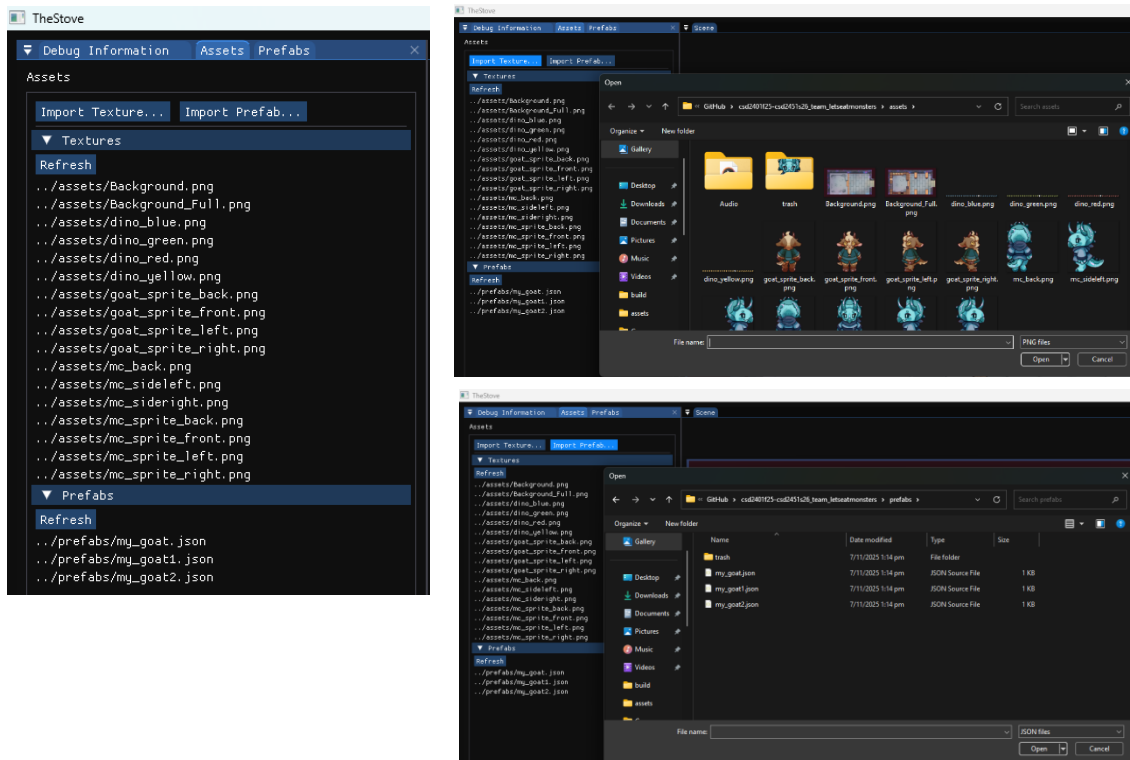




This is the overview of how the ImGui should look like.



Assets Panel manages importing, listing, dragging, and deleting textures or prefabs.

Features & Functions:

Import Texture / Prefab

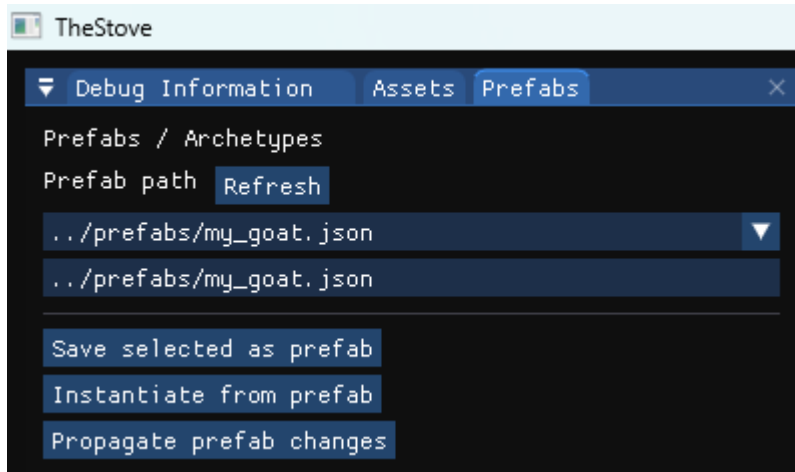
- Opens native file dialogs.
- Copies selected files into project folders (../assets/ or ../prefabs/).
- Auto-refreshes asset lists.
- Optionally applies the imported texture directly to the currently selected object (unless Ctrl is held).

Texture & Prefab Listings

- Displays all available assets using ImGui selectable entries.
- Double-click a texture to apply to the currently selected GameObject.
- Right-click to open context menu for **soft delete** (moves file to /trash).

Drag & Drop

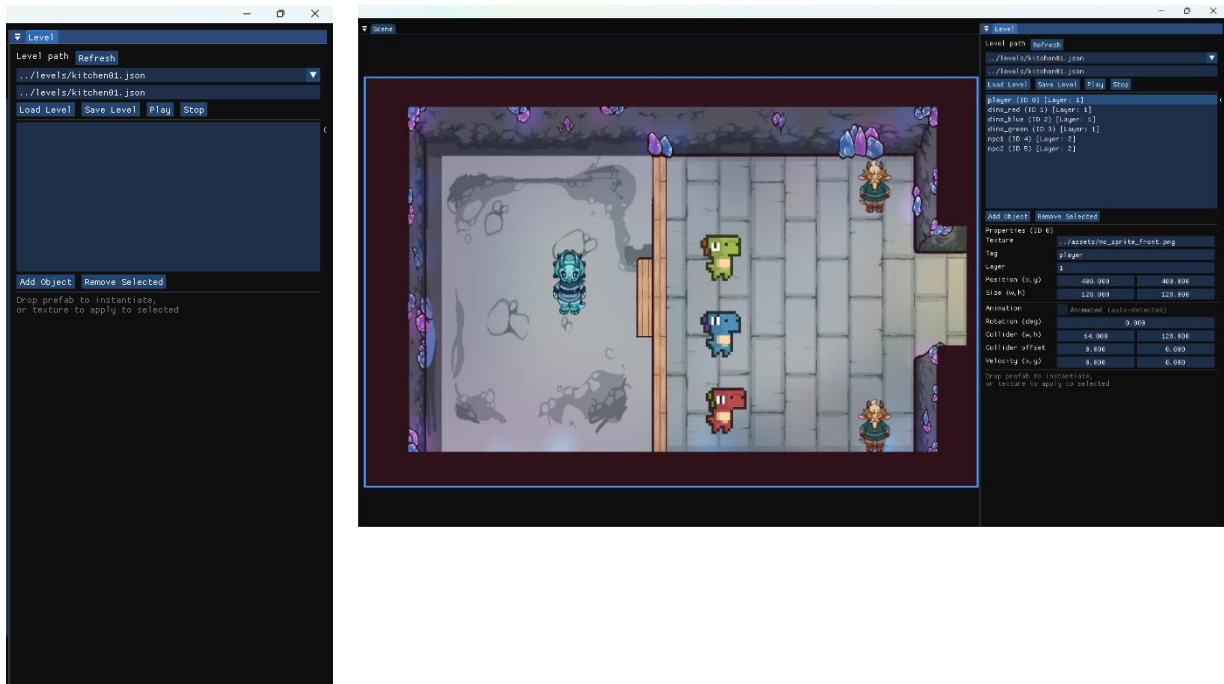
- Starts drag payloads:
 - "ASSET_PATH" (for textures)
 - "PREFAB_PATH" (for prefabs)
- These payloads are received by the Level panel to instantiate or apply textures.



Prefabs Panel saves selected objects as reusable prefabs, instantiates prefabs, and propagates changes to linked instances.

Features & Functions:

- **Path Selection**
 - Dropdown listing all ../prefabs/*.json files.
 - Manual input to set or edit path.
- **Save Selected as Prefab**
 - Converts the selected GameObject into a prefab file.
 - Saves transform, collider, tag, velocity, and animation state.
 - Links that GameObject's ID to the saved prefab.
- **Instantiate from Prefab**
 - Spawns a new GameObject using prefab data (animated or static).
 - Automatically applies:
 - Texture
 - Rotation
 - Collider
 - NPC velocity
 - Tag-based ID assignment (player, npc1, etc.)
 - Links instance to prefab for later propagation.
- **Propagate Changes**
 - Updates all linked instances in the scene when prefab file changes.



Level Panel is the core editor view: loads/saves levels, lists scene objects, edits their properties, and handles play/stop simulation.

Key Functions:

Level File Controls

- Dropdown & text input for level path (./levels/*.json).
- **Load Level:** loads JSON and builds GameObjects in the scene.
- **Save Level:** converts scene state back to LevelData and saves to JSON.
- **Play / Stop:**
 - **Play:** creates a “snapshot,” clears scene, rebuilds in play mode (with physics/logic).
 - **Stop:** restores from snapshot and exits simulation.

Object Hierarchy

- Lists all GameObjects with name, ID, and layer.
- Clicking an object sets it as **selected**, enabling the **Properties Inspector**.

Properties Inspector

- Texture path: Updates sprite + animation logic automatically
- Tag: player, npc1, npc2, dino affect scene logic and IDs
- Layer: Read-only
- Position/Size: Drag to field right-click “Reset to Default”
- Rotation: Stored in degrees
- Collider size/offset: Updates collision boxes
- Velocity: Used for NPC movement when testing
- Animation: Auto-detected; drop-down for available clips if animated

Drag-Drop Integration

At the bottom, a Scene Viewport placeholder (“Drop prefab to instantiate, or texture to apply”):

- Accepts PREFAB_PATH drops and spawns new GameObject from prefab file.
- Accepts ASSET_PATH drops and applies texture to selected GameObject.